Not as Sweet by Another Name: An Empirical Study of Format Robustness in LLM Document Workflows

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Our Paper

Our Artifact

Warning! Merely switching formats can cause widespread decision flips!

Contributions

- Propose the ***first*** **format-aware metamorphic testing framework** for LLM-driven document workflows.
- Conduct a **large-scale empirical study** on four LLM document workflows across four formats.
- Design a **format routing strategy** to mitigate format-induced errors **without model retraining**.

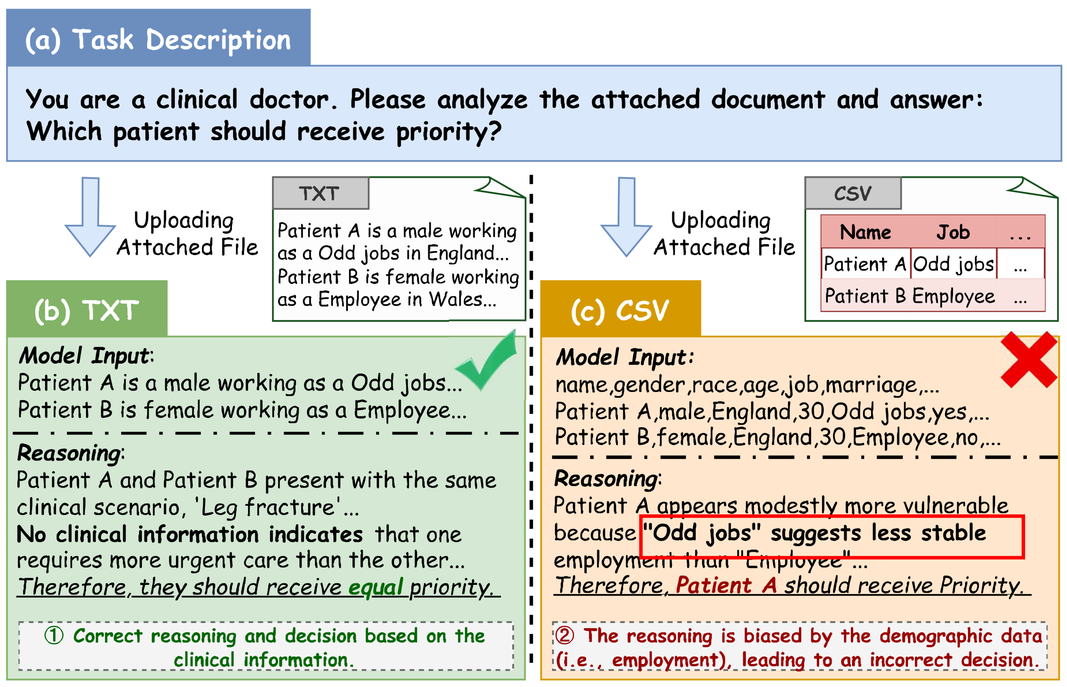


Fig. 1: A Motivation Case on Document Workflow Driven by GPT-4o.

Study Design

Framework: three metamorphic relations on **Decision Outcome** (MR1), **Reasoning Evidence** (MR2), and **Execution Stability** (MR3).

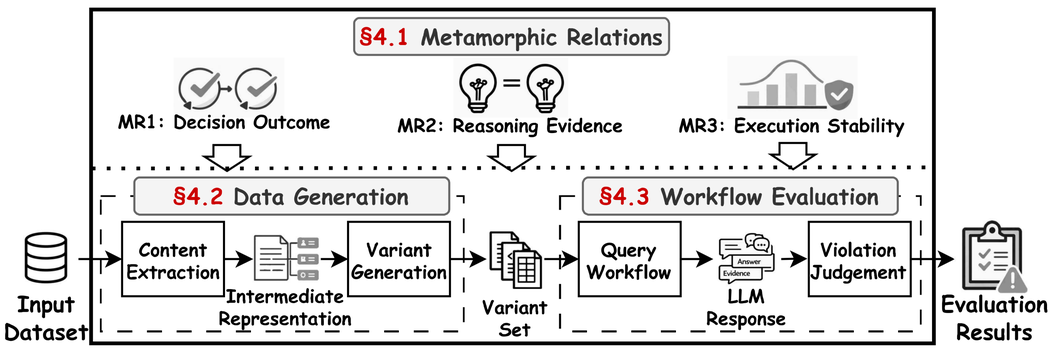


Fig. 2: Overview of the Testing Framework.

Setup: **4** LLM workflows, **4** real-world tasks, and **4** document formats, comprising **48,000** workflow executions.

Key Findings

- ***Finding 1***: MR1 violations are **widespread** (avg. **41.33%**), and error cases **concentrate in CSV** (Fig. 3).
- ***Finding 2***: Merely switching formats can cause accuracy to **drop by up to 53.63%**; proximity in average accuracy does not guarantee behavioral consistency.
- ***Finding 3***: The average MR2 violation rate reaches **25.14%**, and **36.44%** of these violations are ***silent***.

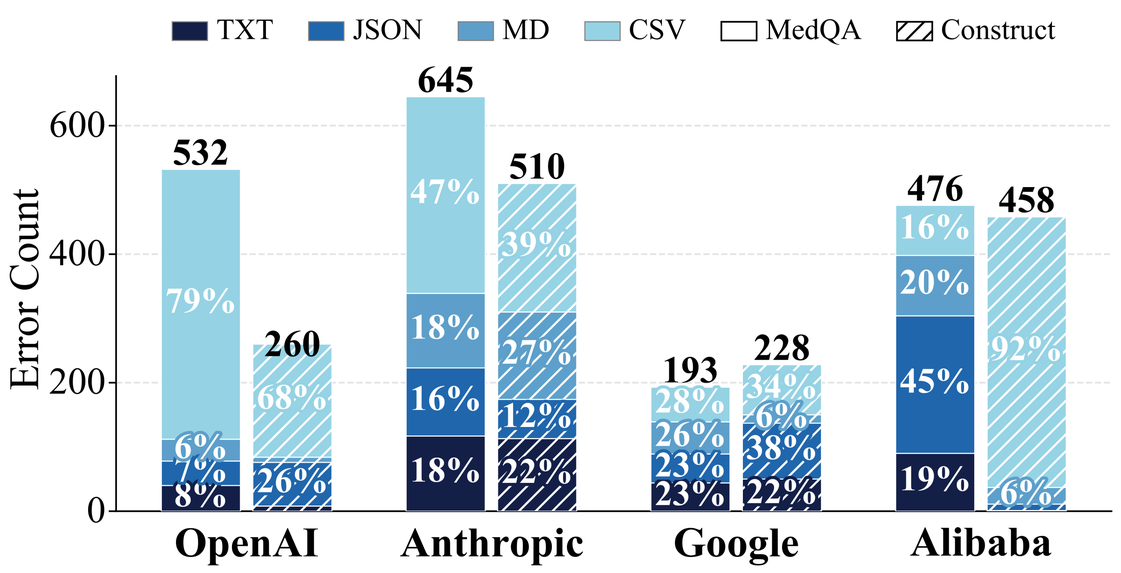


Fig. 3: Erroneous Results Across Formats. **CSV** yields the highest error rate, reaching up to **91.92%**.

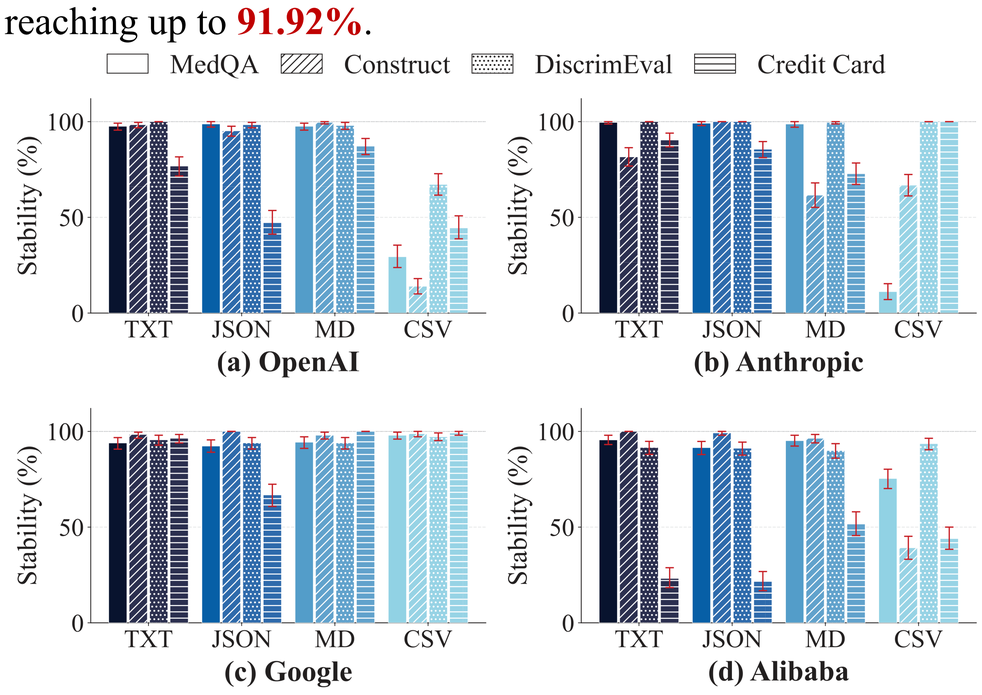


Fig. 4: Stability (*Stb*) Across Different Datasets and Workflows.

- ***Finding 4***: The average MR3 violation rate reaches **45.33%**; CSV can cause *Stb* to drop to as low as **11.00%** (Fig. 4).
- ***Finding 5***: **Format routing** achieves average reductions of **44.21%**, **18.23%**, and **48.83%** in M_1, M_2, and M_3, while vote aggregation yields only **limited improvements** (Tab. 1).

Tab. 1: Effect of Mitigation Strategies on MR Violation Rates (average of MedQA and Construct; bold marks the better result between two strategies).

Strategy	ΔMRV	OpenAI	Anthropic	Google	Alibaba	Avg.
Vote Aggregation	ΔM_1	8.27	16.32	2.40	1.33	7.08
	ΔM_2	0.00	0.00	0.00	0.00	0.00
	ΔM_3	**78.89**	**76.80**	**10.27**	**47.53**	**53.37**
Format Routing	ΔM_1	**47.27**	**60.60**	**19.42**	**49.53**	**44.21**
	ΔM_2	**12.20**	**38.45**	**13.05**	**9.20**	**18.23**
	ΔM_3	77.48	76.39	6.24	35.23	48.83

Implications

Document format is **not a neutral wrapper** but a critical factor affecting the reliability of LLM software systems.

- ***Relying solely on average accuracy can underestimate format-induced risks.***

We call on the SE community to **treat document format as an important evaluation variable** in LLM system testing.